

Linear Regression Model

$$y_t = x_t \beta + u_t, \quad t=1, 2, \dots, N$$

$|x| \quad |x| \times K \quad |x|$

$$y = X\beta + u$$

$N \times 1 \quad N \times K$

- The least square method of estimation of unknown coefficient β finds the vector $\hat{\beta}$ that minimizes the sum of squared residuals (SSR)

$$\min_{\beta} SSR = \sum_{t=1}^N (y_t - x_t \beta)^2 = (y - X\beta)'(y - X\beta)$$

$$\text{FOC} \quad \frac{\partial SSR}{\partial \beta} = 2(-X'y + X'X\hat{\beta}) = 0$$

$$\text{SOC} \quad \frac{\partial^2 SSR}{\partial \beta \partial \beta'} = 2X'X > 0$$

Properties: existence + uniqueness

(A1): $\text{rank}(X) = K$ and $\text{rank}(X, y) = K+1$

↳ (a) $\hat{\beta}$ is unique

(b) $\hat{\beta}$ is a linear function of y

(c) Def: $\hat{u} = y - X\hat{\beta}$

(c.1) $\hat{u} = My = Mu$ where $M = I - X(X'X)^{-1}X'$

(c.2) $\hat{u}$ is orthogonal to the columns of X .

$$X' \hat{u} = \sum_{t=1}^N X_t' \hat{u}_t = 0 \quad \forall t$$

(c.3) If X has a column of constant, $\sum_{t=1}^N \hat{u}_t = 0$

$$X' \hat{u} = 0$$

(A2) $E(u|x) = 0 \Rightarrow E(u) = 0$ (strongly exogenous)

↳ $E(u_t | x_s) = 0 \quad \forall t, s$
 $(E(u_t | x_t) = 0 \quad \forall t \text{ (weakly exogenous)})$

unbiasedness

↳ (d) $\hat{\beta}$ is unbiased

(e) R : $J \times K$ matrix of nonzero constants.
 An unbiased estimator of $R\beta$ is $R\hat{\beta}$.

(f) $E(\hat{u} - u | x) = 0$ ($\hat{u}$ is an unbiased estimator of the error term in the "conditional" sense).

(A3) $E(uu' | x) = \sigma^2 I$ ($E(uu' | x) = \text{Cov}(u, u' | x)$)

- Conditional homoscedasticity

- $E(u_t^2 | x_s) = \sigma^2 \quad \forall t, s$ and

$E(u_t u_{t'} | x_s) = 0 \quad \forall t \neq t', s$

↳ (g) $V \equiv \text{Cov}(\hat{\beta} | x) = \sigma^2 (X'X)^{-1}$

(idea) otherwise,

$$V = E[(X'X)^{-1} X' u u' X (X'X)^{-1}]$$

(h) Gauss-Markov Thm. (PF)

Under (A1)-(A3), $\hat{\beta}_{OLS}$ is BLUE of β

!! C.F

$$V \equiv \text{Cov}(\hat{\beta} | x) = \sigma^2 (X'X)^{-1}$$

$$\text{Cov}(\hat{\beta}) = E_x(\text{Cov}(\hat{\beta} | x)) = \sigma^2 E[(X'X)^{-1}]$$

(i) The unique BLUE of linearity estimable linear parametric functions $R\beta$ is given by $R\hat{\beta}$.

(j) $\text{Cov}(\hat{u} | x) = \sigma^2 M$, which is singular.

$\hat{u}_t$ and $\hat{u}_s$ are correlated even if we assumed that the error terms u_t and u_s are uncorrelated.

(k) $\hat{\sigma}^2 = \frac{SSR}{N-K} = \frac{\hat{u}' \hat{u}}{N-K}$ which is

unbiased estimator of σ^2 .

- the unbiased estimator of $\text{Cov}(\hat{\beta} | x)$ is $\hat{\sigma}^2 (X'X)^{-1}$

- the unbiased estimator of $\text{Cov}(R\hat{\beta} | x)$ is $\hat{\sigma}^2 R(X'X)^{-1}R'$

(A4) $u|x \sim N(0, \sigma^2 I)$

$\hookrightarrow (i) \hat{\beta}|x \sim N(\beta, V)$

$\hat{u}|x \sim N(0, \sigma^2 M)$

(m) $W \equiv \frac{(n-k) \hat{\sigma}^2}{\sigma^2} = \frac{\hat{u}'\hat{u}}{\sigma^2} = \frac{u'Mu}{\sigma^2} \sim \chi^2(N-k)$

$V(n) \hat{\beta} \perp \hat{\sigma}^2$

Aside:

See some statistical properties on p34

(o) $R\hat{\beta}|x \sim N(R\beta, RVR')$

(p) $Z^* \equiv (R\hat{\beta} - R\beta)'(RVR')^{-1}(R\hat{\beta} - R\beta) \sim \chi^2(J)$

(q) $Z \equiv (R\hat{\beta} - q)'(RVR')^{-1}(R\hat{\beta} - q) \sim \chi^2(J, \delta)$

$\delta^2 = (R\beta - q)'(RVR')^{-1}(R\beta - q)$

(r) $F^* = \frac{Z^*/J}{W/(N-k)}$

$= \frac{1}{J} (R\hat{\beta} - R\beta)'(R\hat{V}R')^{-1}(R\hat{\beta} - R\beta) \sim F_{N-k}^J$

(s) $F = \frac{Z/J}{W/(N-k)}$

$= \frac{1}{J} (R\hat{\beta} - q)'(R\hat{V}R')^{-1}(R\hat{\beta} - q) \sim F_{N-k}^J(\delta)$

Remark:

See discussion on t on p5-6

$\hookleftarrow \text{idea} \frac{(N-k) \hat{\sigma}^2}{\sigma^2} \sim \chi^2(N-k)$

$u'Mu = (Mu)'(Mu)$ and $Mu \sim N(0, \sigma^2 M)$

Thus $(Mu)'(\sigma^2 M)^{-1}(Mu) \sim \chi^2$

$\Rightarrow (\sigma^2)^{-1} u'Mu \sim \chi^2$

$\Rightarrow (\sigma^2)^{-1} u'Mu \sim \chi^2$

Measurement of Goodness of Fit

A measurement of the goodness of fit is the squared correlation coefficient b/w the y_t and $\hat{y}_t$.

$$R^2 = [\text{Corr}(y_t, \hat{y}_t)]^2 = \frac{(\text{Cov}(y_t, \hat{y}_t))^2}{\text{Var}(y_t) \text{Var}(\hat{y}_t)}$$

$$= \frac{(y - \bar{y})'(\hat{y} - \bar{\hat{y}})/N}{[(y - \bar{y})'(y - \bar{y})/N][(\hat{y} - \bar{\hat{y}})'(\hat{y} - \bar{\hat{y}})/N]}$$

Remarks

1. $R^2 \in [0, 1]$

2. R^2 is unit-free

Special Case: If a constant is one of the regressors

Lemma

(a) $\bar{y} = \bar{\hat{y}}$ key: $y = \hat{y} + \hat{u}$

(b) $\hat{u}'(\hat{y} - \bar{\hat{y}}) = 0$

(c) $(y - \bar{y})'(\hat{y} - \bar{\hat{y}}) = (\hat{y} - \bar{\hat{y}})'(\hat{y} - \bar{\hat{y}})$

(d) $(y - \bar{y})'(y - \bar{y}) = (\hat{y} - \bar{\hat{y}})'(\hat{y} - \bar{\hat{y}}) + \hat{u}'\hat{u}$

Claim

In this special case,

$$R^2 = \frac{(\hat{y} - \bar{\hat{y}})'(\hat{y} - \bar{\hat{y}})}{(y - \bar{y})'(y - \bar{y})} = 1 - \frac{\hat{u}'\hat{u}}{(y - \bar{y})'(y - \bar{y})}$$

Drawback

A serious drawback of using R^2 as a measure of goodness of fit is that it is nondecreasing, and typically an increasing, in the number of regressors

Adjusted R^2 , $\bar{R}^2$

$$\bar{R}^2 = 1 - \frac{\hat{u}'\hat{u}/n-k}{(y-\bar{y})'(y-\bar{y})/n-1}$$

$$= 1 - \frac{n-1}{n-k} (1 - R^2), \quad \frac{-k+1}{n-k} \leq \bar{R}^2 \leq 1$$

Remark

$\bar{R}^2$ is less than 1, but it can take a negative value if $R^2 < \frac{k-1}{n-k}$

✓ Model Selection : (P8)

$$1 - \bar{R}^2 = \frac{SSR/n-k}{SST/n-1} = \frac{n-1}{SST} \frac{SSR}{n-k}$$

$$\frac{SSR}{n-k} \uparrow \Rightarrow 1 - \bar{R}^2 \downarrow \Rightarrow \bar{R}^2 \uparrow$$

- Akaike Information Criterion

$$AIC: \ln\left(\frac{\hat{u}'\hat{u}}{n}\right) + 2\left(\frac{k}{n}\right) \text{ penalty}$$

- Schwartz Information Criterion

$$SIC: \ln\left(\frac{\hat{u}'\hat{u}}{n}\right) + (\ln n) \frac{k}{n}$$

✓ Thm (BLUP) $\rightarrow$ the variance of prediction error

Consider a linear regression model $y_1 = X_1\beta + u_1$. Suppose we wish to predict the value of dependent variable at a new values of regressor X_2 when the regression relationship has not changed.

i.e. $y_2 = X_2\beta + u_2$ and $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ satisfy EES

(A1) - (A3). Then, the best linear unbiased predictor (BLUP) of y_2 given

X_2 is $\hat{y}_2 = X_2\hat{\beta}$ linear in y_1 unbiased in the sense that the prediction error is zero.

LS E of SLM w/ Non-stochastic Linear Restriction on Coefficients

$$\min \mathcal{L} = (y - X\beta)'(y - X\beta) + 2\lambda'(R\beta - q)$$

FOC :-

$$\frac{\partial \mathcal{L}}{\partial \beta} = -2X'(y - X\beta) + 2R'\lambda = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 2(R\beta - q) = 0$$

$$\Rightarrow \tilde{\beta} = \hat{\beta} - VR'(RVR')^{-1}(R\hat{\beta} - q)$$

$$= \hat{\beta} - VR'S^{-1}(R\hat{\beta} - q)$$

$$V = \sigma^2(X'X)^{-1}$$

$$S = R[\sigma^2(X'X)^{-1}]^{-1}R' = RVR'$$

$$\tilde{\lambda} = \sigma^2(RVR')^{-1}(R\hat{\beta} - q) \quad \text{MLE}$$

$$= \sigma^2 S^{-1}(R\hat{\beta} - q) \quad \text{cov}(\tilde{\theta}) = \Phi^{-1} - \Phi^{-1}H(H'\Phi^{-1}H)^{-1}H'\Phi^{-1}$$

Properties

(a) Moments of $\tilde{\beta} = \hat{\beta} - VR'S^{-1}(R\hat{\beta} - q)$

$$(a.1) E(\tilde{\beta}) = \beta - VR'S^{-1}(R\beta - q)$$

$$(a.2) \tilde{V} = \text{cov}(\tilde{\beta}|x) = V - VR'S^{-1}RV$$

$$(a.3) \text{cov}(\tilde{\beta}|x) - \text{cov}(\hat{\beta}|x) = VR'S^{-1}RV \text{ which is PSD}$$

$$(a.4) \text{MSE}(\tilde{\beta}|x) - \text{MSE}(\hat{\beta}|x)$$

$$\text{MSE}(\tilde{\theta}) = \text{var}(\tilde{\theta}) + [b; a; \tilde{\theta}]^2 = VR'S^{-1}[S - (R\beta - q)(R\beta - q)']S^{-1}RV$$

(b) Moments of $\tilde{\lambda} = \sigma^2 S^{-1}(R\hat{\beta} - q)$

$$(b.1) E(\tilde{\lambda}) = \sigma^2 S^{-1}(R\beta - q)$$

$$(b.2) \text{cov}(\tilde{\lambda}) = \sigma^4 S^{-1}$$

$$(b.3) \text{cov}(\tilde{\beta}, \tilde{\lambda}) = 0$$

Remark:

$$E(\tilde{\lambda}) = 0 \Leftrightarrow R\beta = q$$

$$(c) \tilde{u} = y - x\tilde{\beta}$$

$$\hookrightarrow (c.1) \tilde{u} = y - x\tilde{\beta}$$

$$= \hat{u} + xVR'S^{-1}(R\hat{\beta} - q)$$

$$= \tilde{M}u + xVR'S^{-1}(R\hat{\beta} - q)$$

$$\text{where } \tilde{M} = M + xVR'S^{-1}R(x'x)^{-1}x'$$

$$\text{and } M = I - x(x'x)^{-1}x'$$

$$(c.2) E(\tilde{u}) = xVR'S^{-1}(R\hat{\beta} - q)$$

$$\text{Cov}(\tilde{u}) = \sigma^2 \tilde{M}$$

$$(c.3) \text{Cov}(\tilde{u}, \tilde{\beta}) = 0$$

$$\text{Cov}(\tilde{u}, \tilde{\lambda}) \neq 0$$

central χ^2
if $R\beta = q$

non-central
 χ^2 distribution

$$\star (d) SSR = \tilde{u}'\tilde{u}$$

$$\hookrightarrow (d.1) SSR = SSR_u + \sigma^2(R\hat{\beta} - q)'S^{-1}(R\hat{\beta} - q)$$

$$= u'\tilde{M}u + 2u'\tilde{M}xVR'S^{-1}(R\hat{\beta} - q) + \sigma^2(R\hat{\beta} - q)'S^{-1}(R\hat{\beta} - q)$$

$$(d.2) SSR \geq SSR_u$$

$$(d.3) E(SSR) = \sigma^2(N - K + J) + \sigma^2(R\hat{\beta} - q)'S^{-1}(R\hat{\beta} - q)$$

$$(d.4) \tilde{\sigma}^2 = \frac{SSR}{N - K + J} \text{ Then,}$$

$$E(\tilde{\sigma}^2) = \sigma^2 + \frac{\sigma^2(R\hat{\beta} - q)'S^{-1}(R\hat{\beta} - q)}{N - K + J}$$

Remarks

$$1. (d.2) \Rightarrow SSR \geq SSR_u \text{ a.s.}$$

$\Rightarrow SSR$ is non-decreasing in the number of regressors.

$$2. (d.4) \Rightarrow E(\tilde{\sigma}^2) = \sigma^2 \text{ if } R\hat{\beta} = q \text{ and}$$

$$E(\tilde{\sigma}^2) > \sigma^2 \text{ if } R\hat{\beta} > q$$

$$\checkmark (e) \text{ If } u|x \sim N(0, \sigma^2 I),$$

$$\hookrightarrow (e.1) \tilde{\beta}, \tilde{\lambda}, \text{ and } \tilde{u} \text{ are normally distributed}$$

$$(e.2)$$

$$\frac{(N - K + J)\tilde{\sigma}^2}{\sigma^2} \sim \chi^2(N - K + J; \delta^2)$$

$$\text{where } \delta^2 = (R\hat{\beta} - q)'S^{-1}(R\hat{\beta} - q) \text{ (pf: see p1a)}$$

$$(e.3) \tilde{\beta} \perp \tilde{\lambda}, \tilde{\beta} \perp \tilde{\sigma}^2, \tilde{\lambda} \not\perp \tilde{\sigma}^2$$

Restricted LSE by Reparameterization

Example

$$Y = \begin{pmatrix} x_1 & 0 \\ 0 & x_2 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} + \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

$$\hookrightarrow \text{unrestricted model} \Rightarrow SSR_u$$

$$H_0: \beta_1 = \beta_2$$

$$\begin{pmatrix} x_1 \\ 0 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$$

Note that

$$Y = \begin{pmatrix} x_1 & 0 \\ 0 & x_2 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} + \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \tilde{X}_1\beta_1 + \tilde{X}_2\beta_2 + u$$

$$\beta_1 = \beta_2 \Rightarrow (\tilde{X}_1 + \tilde{X}_2)\beta_1 + u = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \beta_1 + u$$

$$\hookrightarrow \text{restricted model} \Rightarrow SSR_r$$

Hypothesis Testing

1. LR-type test

$\hookrightarrow$ Comparing the values of the objective function.
(ie SSR)

$$\frac{SSR_r - SSR_u}{\sigma^2} = (R\hat{\beta} - q)'(RVR')^{-1}(R\hat{\beta} - q)$$

$$\Rightarrow \frac{(R\hat{\beta} - q)'(R\hat{V}R')^{-1}(R\hat{\beta} - q)}{J} \sim F_{N-K}^J$$

$$\hat{V} = \hat{\sigma}^2(x'x)^{-1}, \quad \hat{\sigma}^2 = \frac{\hat{u}'\hat{u}}{N-K}$$

$$\frac{SSR_r - SSR_u / J}{SSR_u / (N-K)}$$

2. Wald-type Test

↳ Comparing the distance b/w $R\hat{\beta}$ and $\tilde{q}$.
(ie test whether $\hat{\beta}$ satisfies the constraint)

↳ let $P = (RVR')^{-1}$

$P(R\hat{\beta} - r) \sim N(0, I)$ under H_0

Why $R\hat{\beta} - r \sim N(R\beta - r, (RVR')^{-1})$ $\xrightarrow{H_0}$
 $P(R\hat{\beta} - r) \sim N(P(R\beta - r), P(RVR')^{-1}P')$

↳ $(R\hat{\beta} - r)' P (R\hat{\beta} - r) \sim \chi^2(J)$

↳ $(R\hat{\beta} - r)' (RVR') (R\hat{\beta} - r)$

↳ $F = \frac{(R\hat{\beta} - r)' (R\hat{V}R')^{-1} (R\hat{\beta} - r)}{J} \sim F_{N-K, J}^J$

3. Lagrange Multiplier (LM) type test.

↳ Comparing the distance b/w $\tilde{\chi}$ and 0.

↳ $\tilde{\chi} \sim N(\sigma^2 (RVR')^{-1} (R\beta - r), \sigma^4 (RVR')^{-1})$

↳ $\tilde{\chi}' (\sigma^4 (RVR')^{-1})^{-1} \tilde{\chi} = \sigma^{-4} \tilde{\chi}' (RVR') \tilde{\chi}$
 $= \sigma^{-2} \tilde{\chi}' R (X'X)^{-1} R' \tilde{\chi} \sim \chi^2(J)$

(*) $F = \frac{1}{J} \tilde{\sigma}^2 \tilde{\chi}' [R (X'X)^{-1} R'] \tilde{\chi} \sim F_{N-K+J, J}^J$
 $\tilde{\sigma}^2 = \frac{\tilde{u}' \tilde{u}}{N-K+J}$

4. Score type test.

↳ ask whether the restricted estimator $\tilde{\beta}$ satisfies the FOC of the unrestricted least squares estimation. (ie $-X'Y + X'X\tilde{\beta} + R'\tilde{\chi} = 0$?)

FOC: $-X'Y + X'X\tilde{\beta} + R'\tilde{\chi} = 0$

$\Rightarrow g(\tilde{\beta}) + R'\tilde{\chi} = 0$

(*) $\Rightarrow F = \frac{1}{J} \tilde{\sigma}^2 [g(\tilde{\beta})]' (X'X)^{-1} [g(\tilde{\beta})]$
 $\sim F_{N-K+J, J}^J$

Nonlinear Hypothesis and Delta Method

Delta Method:

$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \Sigma)$

$h(\beta)$: $J \times 1$ restriction

$\frac{\partial h(\beta)}{\partial \beta} : K \times J$

$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} \left(\frac{\partial h(\beta)}{\partial \beta} \Big|_{\beta=\hat{\beta}} \right)' N(0, \Sigma)$
 $= N(0, \left(\frac{\partial h(\beta)}{\partial \beta} \Big|_{\beta=\hat{\beta}} \right)' \Sigma \frac{\partial h(\beta)}{\partial \beta} \Big|_{\beta=\hat{\beta}})$

Bootstrap Method

When the normality assumption is invalid or the distribution of error term is unknown, or the hypothesis is nonlinear, we typically rely on the asymptotic justification.

The bootstrap method is more commonly used for computing the standard errors and/or confidence intervals of $\hat{\beta}$ (or $h(\hat{\beta})$)

Step 1: RUN OLS to get $\hat{\beta}$ (or $h(\hat{\beta})$) and $\hat{u}$

Step 2: Construct a new residual vector $\tilde{u}$ by random selecting elements from $\hat{u}$ with replacement, and new bootstrap sample

$$\tilde{y}_t = x_t \hat{\beta} + \tilde{u}_t$$

Step 3: Compute $\tilde{\beta}$ and $h(\tilde{\beta})$

Step 4: Repeat step 2 and step 3 and Collect $\tilde{\beta}$ and $h(\tilde{\beta})$

Step 5: Compute the empirical standard errors and confidence interval from the collected $\tilde{\beta}$ and $h(\tilde{\beta})$.

MLE

Notation

$$1. L(y; \theta) = \prod_{t=1}^N f(y_t; \theta)$$

$$\begin{aligned} \ell(y; \theta) &= \ln L(y; \theta) = \sum_{t=1}^N \ln f(y_t; \theta) \\ &= \sum_{t=1}^N \ell(y_t; \theta) \end{aligned}$$

$$2. g_t(\theta) = \frac{\partial \ell(y_t; \theta)}{\partial \theta}$$

$$g(\theta) = \frac{\partial \ell(y; \theta)}{\partial \theta} = \sum_{t=1}^N \frac{\partial \ell(y_t; \theta)}{\partial \theta} = \sum_{t=1}^N g_t(\theta)$$

$$H_t(\theta) = \frac{\partial^2 \ell(y_t; \theta)}{\partial \theta \partial \theta'}$$

$$H(\theta) = \frac{\partial^2 \ell(y; \theta)}{\partial \theta \partial \theta'} = \sum_{t=1}^N \frac{\partial^2 \ell(y_t; \theta)}{\partial \theta \partial \theta'} = \sum_{t=1}^N H_t(\theta)$$

Lemma

$$1. E\left(\frac{\partial \ell(\theta_0)}{N}\right) \geq E\left(\frac{\partial \ell(\theta)}{N}\right) \quad \forall \theta \in \Theta$$

$$2. \text{plim} \frac{\partial \ell(\theta)}{N} = E\left(\frac{\partial \ell(\theta)}{N}\right)$$

$$3. E(g(\theta)) = E(g_t(\theta)) = 0$$

$$\text{cov}(g(\theta)) = N\Phi, \quad \text{cov}(g_t(\theta)) = \Phi$$

$$\Phi = E\left(-\frac{1}{N} H(\theta)\right)$$

$$4. \frac{1}{\sqrt{N}} g(\theta) \xrightarrow{d} N(0, \Phi) \quad \text{notice: NO INVERSE}$$

$$5. \text{plim} \left(-\frac{1}{N} H(\theta)\right) = \Phi$$

Thm

Let $\hat{\theta}$ be the MLE of the true parameter θ_0 for a sample of size n

$$(a) \hat{\theta} \xrightarrow{P} \theta_0$$

$$(b) \sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \Phi^{-1}) \quad \text{from ipdf}$$

$$\begin{aligned} \text{where } \Phi &= E\left(-\frac{1}{N} H(\hat{\theta})\right) \\ &= E(-H_t(\theta)) \\ &= E(g_t(\theta) g_t(\theta)'). \end{aligned}$$

Restricted MLE

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(\mu_{\hat{\theta}}, \Sigma_{\hat{\theta}})$$

$$\frac{1}{\sqrt{n}} \tilde{\chi} \xrightarrow{d} N(\mu_{\tilde{\chi}}, \Sigma_{\tilde{\chi}}) \quad H = \frac{\partial H(\theta)}{\partial \theta} \bigg|_{\theta=\theta_0}$$

$$\mu_{\hat{\theta}} = \Phi^{-1} H (H' \Phi^{-1} H)^{-1} h(\theta_0) \quad \text{restriction}$$

$$\Sigma_{\hat{\theta}} = \Phi^{-1} - \Phi^{-1} H (H' \Phi^{-1} H)^{-1} H' \Phi^{-1}$$

$$\mu_{\tilde{\chi}} = - (H' \Phi^{-1} H)^{-1} h(\theta_0)$$

$$\Sigma_{\tilde{\chi}} = (H' \Phi^{-1} H)^{-1}$$

$$\langle \text{Key} \rangle \ell(\theta) - \tilde{\chi}' h(\theta)$$

$$\Rightarrow g(\tilde{\theta}) - H(\tilde{\theta}) \tilde{\chi} = 0 \text{ and } h(\tilde{\theta}) = 0$$

$$\Rightarrow \text{Taylor Expansion}$$

$$\Rightarrow \begin{bmatrix} -\frac{1}{N} H_0 & H_0 \\ H_0' & 0 \end{bmatrix} \begin{bmatrix} \sqrt{n}(\hat{\theta} - \theta_0) \\ \frac{1}{\sqrt{n}} \tilde{\chi} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{n}} g(\theta_0) \\ -\sqrt{n} h(\theta_0) \end{bmatrix}$$

$$N \left(\begin{pmatrix} 0 \\ -\sqrt{n} h(\theta_0) \end{pmatrix}, \begin{pmatrix} \Phi & 0 \\ 0 & 0 \end{pmatrix} \right)$$

Asymptotic Tests $H_0: \beta(0)=0, H_1: \beta(0) \neq 0$

I. Likelihood Ratio Test (LRT)

: To compare the objective function

$$\Rightarrow LRT = 2(\ln L(\hat{\theta}) - \ln L(\tilde{\theta}))$$

$$LRT \xrightarrow{d} \chi^2(J)$$

II. Wald Test (W)

: To compare the values of the restriction equation evaluated at unrestricted and restricted estimators

$$\Rightarrow W = N h(\hat{\theta})' (H' \Psi^{-1} H)^{-1} h(\hat{\theta}) \xrightarrow{d} \chi^2(J)$$

PF (by delta method)

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \Psi^{-1})$$

$$\Rightarrow \sqrt{n}(h(\hat{\theta}) - h(\theta)) \xrightarrow{d} \left(\frac{\partial h}{\partial \theta}\right)' N(0, \Psi^{-1})$$

$$\xrightarrow{d} N(0, H' \Psi^{-1} H)$$

III. Lagrange Multiplier Test (LM) $h(\hat{\theta})=0$

: To compare the Lagrange multiplier and 0.

$$\Rightarrow LM = \frac{1}{N} \tilde{\lambda}' (H' \Psi^{-1} H) \tilde{\lambda} \xrightarrow{d} \chi^2(J)$$

$$\text{PF: } \frac{1}{\sqrt{n}} \tilde{\lambda} \xrightarrow{d} N(-(H' \Psi^{-1} H)^{-1} h(\theta_0), (H' \Psi^{-1} H)^{-1})$$

Remark:

$$LM = \frac{1}{N} \tilde{\lambda}' (H' \Psi^{-1} H) \tilde{\lambda}$$

$$= \frac{1}{N} (H \tilde{\lambda})' \Psi^{-1} (H \tilde{\lambda}) = \frac{1}{N} (g_{\lambda}(\theta))' \Psi^{-1} g_{\lambda}(\theta)$$

$\Rightarrow$ SCORE Test

Recall that

$$\frac{1}{\sqrt{n}} g_{\lambda}(\theta) \xrightarrow{d} N(0, \Psi)$$

PDF Editor

MLE of Standard Linear Regression Model

$$H = \frac{\partial^2 \ln l(\theta)}{\partial \theta \partial \theta'} = \begin{bmatrix} -\frac{1}{\sigma^2} X'X & -\frac{1}{\sigma^4} X'u \\ -\frac{1}{\sigma^4} X'u & \frac{N}{2\sigma^4} - \frac{u'u}{\sigma^6} \end{bmatrix}$$

$$\Rightarrow \Psi = E(-\frac{1}{N} H) = \begin{bmatrix} \frac{1}{\sigma^2} \frac{1}{N} E(X'X) & 0 \\ 0 & \frac{1}{2\sigma^4} \end{bmatrix}$$

Thus,

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \Psi^{-1}) = N(0, \begin{bmatrix} \sigma^2 N E(X'X)^{-1} & 0 \\ 0 & 2\sigma^4 \end{bmatrix})$$

$$\Rightarrow \text{COV} \begin{pmatrix} \hat{\beta}_{MLE} \\ \hat{\sigma}_{MLE}^2 \end{pmatrix} = \begin{bmatrix} \sigma^2 E(X'X)^{-1} & 0 \\ 0 & \frac{2\sigma^4}{N} \end{bmatrix}$$

Remark:

plug $\hat{\beta}_{MLE}, \hat{\sigma}_{MLE}^2$ back to the objective function,

$$l(\hat{\theta}) = \frac{n}{2} \ln 2\pi - \frac{n}{2} \ln \hat{\sigma}_{MLE}^2 - \frac{1}{2} \frac{(y - X\hat{\beta})'(y - X\hat{\beta})}{\hat{\sigma}_{MLE}^2}$$

$$= \frac{n}{2} \ln 2\pi - \frac{n}{2} \ln \hat{\sigma}_{MLE}^2 - \frac{1}{2} \frac{n \hat{\sigma}_{MLE}^2}{\hat{\sigma}_{MLE}^2}$$

$$= \frac{n}{2} \ln 2\pi - \frac{n}{2} \ln \hat{\sigma}_{MLE}^2 - \frac{n}{2} \quad (*)$$

Restricted MLE of Standard Linear Regression

$$\text{Max } l(\theta) = -X'(R\beta - q)$$

Foc:

$$\frac{\partial l}{\partial \beta} = \frac{1}{\sigma^2} (X'y - X'X\tilde{\beta}) - R'\tilde{\lambda} = 0$$

$$\frac{\partial l}{\partial \sigma^2} = -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4} (y - X\tilde{\beta})'(y - X\tilde{\beta}) = 0$$

$$\frac{\partial l}{\partial \lambda} = R\tilde{\beta} - q = 0$$

Remark:

$$\text{MLE-2 } \hat{\sigma}_{MLE}^2 = \frac{\tilde{u}'\tilde{u}}{N} \neq \frac{\tilde{u}'\tilde{u}}{N-K+J} = \hat{\sigma}_{OLS}^2$$

Thus,

$$\begin{pmatrix} \tilde{\beta} \\ \tilde{\lambda} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sigma^2} X'X & R' \\ R & 0 \end{pmatrix}^{-1} \begin{pmatrix} \frac{1}{\sigma^2} X'Y \\ q \end{pmatrix}$$

$$= \begin{pmatrix} \hat{\beta} - VR'S^{-1}(R\hat{\beta} - q) \\ \underbrace{S^{-1}(R\hat{\beta} - q)} \end{pmatrix} \quad \begin{matrix} V = \sigma^2 (X'X)^{-1} \\ S = RVR' \\ = S^{-1}(RVR'S^{-1}R' - q) \end{matrix}$$

Remark $\rightarrow$ different from $\tilde{\lambda}_{OLS}$!!!

$$\tilde{\beta}_{MLE} = \tilde{\beta}_{OLS} ; \tilde{\lambda}_{MLE} \neq \tilde{\lambda}_{OLS} = \sigma^2 S^{-1}(R\hat{\beta} - q)$$

Similarly

$$\text{Cov} \begin{pmatrix} \sqrt{n}(\tilde{\theta} - \theta_0) \\ \frac{1}{\sqrt{n}} \tilde{\lambda} \end{pmatrix} = \begin{pmatrix} \sigma^2 M_{XX}^{-1} - \sigma^2 M_{XX}^{-1} R' (\sigma^2 R M_{XX}^{-1} R')^{-1} \sigma^2 R M_{XX}^{-1} & 0 & 0 \\ 0 & 2\sigma^4 & 0 \\ 0 & 0 & (\sigma^2 R M_{XX}^{-1} R')^{-1} \end{pmatrix}$$

$$\Rightarrow \text{Cov} \begin{pmatrix} \tilde{\beta} \\ \tilde{\sigma}^2 \\ \tilde{\lambda} \end{pmatrix} = \begin{pmatrix} V - VR'S^{-1}RV & 0 & 0 \\ 0 & \frac{2\sigma^4}{N} & 0 \\ 0 & 0 & S^{-1} \end{pmatrix} \quad \begin{matrix} \text{where} \\ M_{XX} = \frac{1}{N} E(X'X) \end{matrix}$$

$$H = \frac{\partial \ell}{\partial \theta} = \begin{bmatrix} K' \\ 0 \end{bmatrix}_{(K+1) \times J}$$

$$\begin{bmatrix} \sigma^2 N E(X'X)^{-1} & 0 \\ 0 & 2\sigma^4 \end{bmatrix}$$

$$\Psi^{-1} - \Psi^{-1} H (H' \Psi^{-1} H)^{-1} H' \Psi^{-1}$$

$$\downarrow \text{Recall} \\ (H' \Psi^{-1} H)^{-1}$$

Remark

Again, plug $\tilde{\beta}$, $\tilde{\sigma}^2$ back to objective fn.

$$\ell(\tilde{\theta}) = \frac{n}{2} \ln \pi - \frac{n}{2} \ln \tilde{\sigma}_{MLE}^2 - \frac{n}{2} \quad (**)$$

Tests of Linear Restriction in Standard Linear Regression Model

I LR Test

$\Rightarrow$ From (*) and (**)

$$\begin{aligned} \text{LRT} &= 2(\ell(\hat{\theta}) - \ell(\tilde{\theta})) \\ &= n(\ln \hat{\sigma}_{MLE}^2 - \ln \tilde{\sigma}_{MLE}^2) \\ &= n \ln \frac{\hat{\sigma}_{MLE}^2}{\tilde{\sigma}_{MLE}^2} \xrightarrow{d} \chi^2(J) \end{aligned}$$

MLE-3

Remark:

$$\begin{aligned} \text{LRT} &= n \left(\ln \frac{\hat{\sigma}_{MLE}^2}{\hat{\sigma}_{MLE}^2} \right) \approx n \left(\frac{\hat{\sigma}_{MLE}^2}{\hat{\sigma}_{MLE}^2} - 1 \right) \\ &= n \frac{\hat{\sigma}_{MLE}^2 - \hat{\sigma}_{MLE}^2}{\hat{\sigma}_{MLE}^2} = n \frac{SSR - SSR_u}{SSR_u} \end{aligned}$$

II. Wald Test

$$W = N \hat{h}(\hat{\theta})' (H' \Psi^{-1} H)^{-1} \hat{h}(\hat{\theta}) \xrightarrow{d} \chi^2(J)$$

where $\hat{h}(\hat{\theta}) = R\hat{\beta} - q$,

$$H = \frac{\partial \hat{h}(\hat{\theta})}{\partial \beta} = \begin{pmatrix} R' \\ 0 \end{pmatrix}$$

$$\Psi^{-1} = \begin{bmatrix} \sigma^2 N E(x'x)^{-1} & 0 \\ 0 & 2\sigma^4 \end{bmatrix}$$

$$\Rightarrow W = (R\hat{\beta} - q)' (R\hat{\sigma}^2 E(x'x)^{-1} R') (R\hat{\beta} - q) \xrightarrow{d} \chi^2(J)$$

III LM Test (see p19-20)

$$\begin{aligned} \text{LMT} &= \frac{1}{N} \tilde{\lambda}' (H' \Psi^{-1} H) \tilde{\lambda} \\ &= \tilde{\lambda}' H' (N \Psi)^{-1} H \tilde{\lambda} \\ &= (R\hat{\beta} - q)' S^{-1} H' (N \Psi)^{-1} H S^{-1} (R\hat{\beta} - q) \\ &= (R\hat{\beta} - q)' \tilde{S}^{-1} (R\hat{\beta} - q) \end{aligned}$$

Remark:

$$\begin{aligned} \text{LMT} &= \frac{1}{N} \tilde{\lambda}' (H' \Psi^{-1} H) \tilde{\lambda} \\ &= \tilde{\lambda}' (H' (N \Psi)^{-1} H) \tilde{\lambda} \\ &= g(\tilde{\theta})' (N \Psi)^{-1} g(\tilde{\theta}) \end{aligned}$$

where $g(\tilde{\theta}) = \begin{pmatrix} \frac{x' \tilde{u}}{\hat{\sigma}^2} \\ 0 \end{pmatrix}$

Thus,

$$\begin{aligned} \text{LMT} &= \frac{\tilde{u}' x (x' x)^{-1} x' \tilde{u}}{\hat{\sigma}^2} \xrightarrow{d} \chi^2(J) \\ &= \frac{\tilde{u}' x (x' x)^{-1} x' \tilde{u}}{\frac{\tilde{u}' \tilde{u}}{N}} \\ &= N \frac{(p_x \tilde{u})' (p_x \tilde{u})}{\tilde{u}' \tilde{u}} \\ &= N \frac{\hat{\tilde{u}}' \hat{\tilde{u}}}{\tilde{u}' \tilde{u}} = N \cdot R_o^2 \end{aligned}$$

where $\hat{\tilde{u}} = \text{from } \tilde{u} \text{ on } X$

Maximum Empirical Likelihood Estimation more than one constraint, $h(\theta)$ m x 1

Suppose that we don't have the complete information contained in the distribution function, but have only info about

moments. Can we still apply the same optimization procedure, replacing the density function w/ moments conditions as constraints?

$$\max_{\theta} \sum_{t=1}^n \ln p_t - K(\sum p_t - 1) + \lambda(\sum p_t h_t(\theta))$$

$$\max_{\theta} \left(\frac{1}{n} \sum \ln p_t - K(\sum p_t - 1) + \lambda(\sum p_t h_t(\theta)) \right)$$

$$\{p_t\} \quad \frac{1}{n} \frac{1}{p_t} - K + \lambda' h_t(\theta) = 0 \quad \text{--- ①}$$

$$\{K\} \quad \sum p_t = 1$$

$$\{\lambda\} \quad \sum p_t h_t(\theta) = 0$$

$$\text{①} \Rightarrow \frac{1}{n} - p_t K + \lambda' p_t h_t(\theta) = 0$$

$$\text{Sum} \Rightarrow 1 - K = 0 \Rightarrow K = 1$$

Focs:

$$\{p_t\} \quad \frac{1}{p_t} - K + \lambda(y_t - \theta) = 0 \quad \text{--- ①}$$

$$\{K\} \quad \sum p_t = 1$$

$$\{\lambda\} \quad \sum p_t (y_t - \theta) = 0$$

$$\text{①} \Rightarrow 1 - p_t K - \lambda p_t (y_t - \theta) = 0$$

$$\text{Sum} \Rightarrow n = K$$

Note:

$$\max \sum \ln p_t$$

$$\Leftrightarrow$$

$$\min \sum \ln(n - \lambda(y_t - \theta))$$

$$\text{s.t. } \sum \frac{y_t - \theta}{n - \lambda(y_t - \theta)} = 0$$

$$\lambda \in \arg \left(\sum \frac{y_t - \theta}{n - \lambda(y_t - \theta)} = 0 \right)$$

$$\hat{\theta} \in \arg \min_{\theta} \max_{\lambda} \sum \ln(n - \lambda' h_t(\theta))$$

Remark:

$$\sqrt{n}(\hat{\theta}_{MLE} - \theta) \xrightarrow{d} N(0, Q^{-1})$$

$$Q = E(g) [E(hh')]^{-1} E(g')$$

$$g = \frac{\partial h}{\partial \theta} = \frac{\partial}{\partial \theta} \sum_{t=1}^n h(y_t, \theta) = \sum_{t=1}^n g_t$$

$$h = h(y, \theta) = \sum_{t=1}^n h(y_t, \theta)$$

$$\hat{g} = \sum_{t=1}^n \hat{p}_t \hat{g}_t, \quad E(hh') = \sum_{t=1}^n \hat{p}_t h(y_t, \hat{\theta}) h(y_t, \hat{\theta})'$$

M.E.L.E. = 1

MEEL

entropy Shannon's entropy

$$\max \sum_{t=1}^n p_t \ln \left(\frac{1}{p_t} \right)$$

$$\text{s.t. } \sum p_t = 1, \sum p_t h_{t+1}(0) = 0$$

$$\Rightarrow \max \sum_{t=1}^n -p_t \ln p_t - K(\sum p_t - 1) + \lambda(\sum p_t h_{t+1}(0))$$

$$\{p_t, \lambda\} \quad \left(-\ln p_t - p_t \frac{1}{p_t} \right) - K + \lambda h_{t+1}(0) = 0 \quad (1)$$

$$\{p_t\} \quad \sum p_t = 1$$

$$\{\lambda\} \quad \sum p_t h_{t+1}(0) = 0 \quad (2)$$

$$(1) \Rightarrow p_t = e^{-K-1} e^{\lambda h_{t+1}(0)} = \frac{e^{\lambda h_{t+1}(0)}}{e^{K+1}}$$

$$\sum p_t = \sum \frac{e^{\lambda h_{t+1}(0)}}{e^{K+1}} = 1$$

$$\Rightarrow e^{K+1} = \sum e^{\lambda h_{t+1}(0)}$$

$$\text{Thus, } p_t = \frac{e^{\lambda h_{t+1}(0)}}{\sum e^{\lambda h_{t+1}(0)}}$$

plug into (2),

$$\sum \frac{e^{\lambda h_{t+1}(0)} h_{t+1}(0)}{\sum e^{\lambda h_{t+1}(0)}} = 0$$

Thus

wondershare™
PDF Editor

MEEL-2

(RE) (or Kullback-Leibler Info Criterion)

$$\max \sum_{t=1}^n p_t \left[\ln \left(\frac{1}{q_t} \right) - \ln \left(\frac{1}{p_t} \right) \right] \quad \text{KL} \quad \ln \frac{p_t}{q_t}$$

$$\text{s.t. } \sum p_t = 1, \sum p_t h_{t+1}(0) = 0$$

$$= \ln n - \sum p_t \ln \frac{1}{p_t}$$

$$\min \text{KL} \left(p, \frac{1}{n} \right) \Leftrightarrow \text{MEEL}$$

$$\text{KL} \left(\frac{1}{n}, q \right) = \sum \frac{1}{n} \left(\ln \frac{1}{q} - \ln n \right)$$

$$= \frac{1}{n} \sum \ln q - \ln n \quad \langle \text{Notes} \rangle$$

$$\min \text{KL} \left(\frac{1}{n}, q \right) \Leftrightarrow \text{MEEL}$$

Continuous distribution
 $f_1(x)$ and $f_2(x)$,

KLIC is

$$\int \ln \frac{f_1(x)}{f_2(x)} \cdot f_1(x) dx$$

(CR) Cressie-Read

Nothing but Box-Cox transformation

$$\max_{p, d} \sum_{t=1}^n p_t \frac{\left(\frac{p_t}{q_t} \right)^d - 1}{d(1+d)}$$

CR

$$\text{CR} \left(p, \frac{1}{n}, d \right) = \sum_{t=1}^n \frac{p_t (np_t)^d - 1}{d(1+d)}$$

Generalization of MEEL

GEL

$$\hat{\theta}_{\text{GEL}} = \underset{\theta}{\operatorname{argmin}} \sup_{\lambda} \sum \psi(x' h_{t|\theta})$$

The GEL considers any objective function $\psi(v_i)$ which is convex in
 v_i in a open interval containing 0
and satisfies the normalization
conditions: $\psi'(0) = -1$, $\psi''(0) = -1$.

$$\hat{\rho} = \frac{\psi_1(x' h_{t|\theta})}{\sum \psi_1(x' h_{t|\theta})}$$

(CUE): $\psi(x' h_{t|\theta}) = -\frac{(1 + x' h_{t|\theta})^2}{2}$

Method of Moments Estimator

$$y_t = \varepsilon_t + \alpha \varepsilon_{t-1}, \quad \varepsilon_t \stackrel{iid}{\sim} (0, \sigma^2)$$

$$E(y_t) = 0,$$

$$E(y_t^2) = \text{Var}(y_t) = \text{Cov}(y_t, y_t)$$

$$= \text{Cov}(\varepsilon_t + \alpha \varepsilon_{t-1}, \varepsilon_t + \alpha \varepsilon_{t-1})$$

$$= (1 + \alpha^2) \sigma^2$$

$$E(y_t y_{t+1}) = \text{Cov}(\varepsilon_t + \alpha \varepsilon_{t-1}, \varepsilon_{t+1} + \alpha \varepsilon_t) = \alpha \sigma^2$$

Thus,

$$\frac{1}{T} \sum y_t^2 = (1 + \alpha^2) \sigma^2$$

$$\frac{1}{T} \sum y_t y_{t+1} = \alpha \sigma^2 \Rightarrow \frac{\alpha}{1 + \alpha^2} = \frac{\sum y_t y_{t+1}}{\sum y_t^2} = \hat{\rho}$$

$$\Rightarrow \hat{\rho} \alpha^2 - \alpha + \hat{\rho} = 0 \Rightarrow \text{get } \hat{\alpha} \Rightarrow \text{get } \hat{\sigma}^2$$

Identifiability:

Parameter θ_0 is said to be identifiable by the population moment condition if and only if $\mu(\theta) = 0$ has a unique solution for θ and the solution is equal to the true value θ_0 .

$$\mu(\theta) = 0 = E(x'(y - z\theta))$$

$$= E(x'(y - z\theta - z\theta_0 + z\theta_0))$$

$$= E(x'(y - z\theta_0)) + E(x'z(\theta_0 - \theta))$$

$$= E(x'z(\theta_0 - \theta)) = E(x'z)(\theta_0 - \theta) = \Sigma_{xz}(\theta_0 - \theta)$$

Thus, the parameter θ is identifiable if

and only if Σ_{xz} has full column rank

That is $\text{Rank}(\Sigma_{xz}) = p$.

Lemma:

The necessary condition (order condition) for the identifiability of θ_0 by the population moment condition is $k \geq p$.

The necessary and sufficient condition (rank condition) for the identifiability of θ_0 by the population moment condition is $\text{rank}(\Sigma_{xz}) = p$.

Overidentification case ($k > p$)

- (i) reduce the number of moment condition to the number of just identified case
- (ii) find an estimator that makes a certain measure of the distance of $m(\theta)$ from zero as small as possible

$$(i) \quad F: k \times p$$

$\Rightarrow$ select p eqn from k eqn

$$\frac{1}{T} F' x' (y - z\theta) = 0$$

$$\therefore \hat{\theta} = (F' x' z)^{-1} F' x' y$$

$$= \theta_0 + (F' x' z)^{-1} F' x' u$$

How to choose F ? To make

$\hat{\theta}$ efficient

$$\text{Cov}(\hat{\theta}) = (F' x' z)^{-1} F' x' \Omega x x' F (F' x' z)^{-1}$$

$$\Rightarrow \min_F \text{Cov}(\hat{\theta}) \Rightarrow F = (x' \Omega x)^{-1} x' z$$

Thus

$$\hat{\theta} = \left(Z'X(X'\Omega X)^{-1}X'Z \right)^{-1} Z'X(X'\Omega X)^{-1}X'Y$$

Remarks.

(a). $Y = Z\theta + u$

$\Rightarrow$ premultiply $X'Y = X'Z\theta + X'u$ $\text{Cov}(X'u) = X'\Omega X$

$\Rightarrow$ GLS $\hat{\theta} = \left(Z'X(X'\Omega X)^{-1}X'Z \right)^{-1} Z'X(X'\Omega X)^{-1}X'Y$

(b) If $\Omega = I$, then $\hat{\theta}_{GLS} = \hat{\theta}_{2SLS}$

Step 1: RUN OLS Z on X

$$Z = X\delta + \varepsilon$$

$$\Rightarrow \hat{\delta} = (X'X)^{-1}X'Z$$

$$\Rightarrow \hat{Z} = X(X'X)^{-1}X'Z$$

Step 2: RUN OLS Y on $\hat{Z}$

$$\hat{\theta} = (\hat{Z}'\hat{Z})^{-1}\hat{Z}'Y$$

$$= (Z'X(X'X)^{-1}X'X(X'X)^{-1}X'Z)^{-1} Z'X(X'X)^{-1}X'Y \quad (2) \sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, V)$$

$$= (Z'X(X'X)^{-1}X'Z)^{-1} Z'X(X'X)^{-1}X'Y$$

(ii)

$$J(\theta) = T \left(\frac{1}{T} X'(Y - Z\theta) \right)' W \left(\frac{1}{T} X'(Y - Z\theta) \right) \\ = \frac{1}{T} (X'(Y - Z\theta))' W (X'(Y - Z\theta))$$

$$\min_{\theta} J(\theta) \quad \left(\frac{\partial J(\theta)}{\partial \theta} \right)$$

$$\Rightarrow Z'XW(X'(Y - Z\theta)) = 0$$

$$\Rightarrow Z'XW X'Y - Z'XW X'Z\theta = 0$$

$$\Rightarrow \hat{\theta} = (Z'XW X'Z)^{-1} Z'XW X'Y$$

$$= \theta_0 + (Z'XW X'Z)^{-1} Z'XW X'u$$

$$\textcircled{1} \hat{\theta} \xrightarrow{p} \theta_0$$

$$\stackrel{pf}{=} (Z'XW X'Z)^{-1} Z'XW X'u$$

$$= \left(\frac{Z'X}{T} W \frac{X'Z}{T} \right)^{-1} \frac{Z'X}{T} W \frac{X'u}{T}$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$\Sigma_{ZX} \quad W_0 \quad \Sigma_{XZ} \quad \Sigma_{ZX} \quad W_0 \quad 0$$

$$\sqrt{T}(\hat{\theta} - \theta_0) = \left(\frac{Z'X}{T} W \frac{X'Z}{T} \right)^{-1} \frac{Z'X}{T} W \frac{X'u}{T} \sqrt{T}$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$\left(\Sigma_{ZX} W_0 \Sigma_{XZ} \right)^{-1} \Sigma_{ZX} W_0 \frac{1}{\sqrt{T}} \sum_{i=1}^T X_i' u_i$$

where $S_0 = \text{plim} \frac{X'u u' X}{T}$

$$V = \left(\Sigma_{ZX} W_0 \Sigma_{XZ} \right)^{-1} \Sigma_{ZX} W_0 S_0 W_0 \Sigma_{XZ} \left(\Sigma_{ZX} W_0 \Sigma_{XZ} \right)^{-1}$$

Thus $V_e = \left(\Sigma_{ZX} S_0^{-1} \Sigma_{XZ} \right)^{-1}$

MME-2

when $W_0 = S_0^{-1} = \left[\text{plim} \left(\frac{X'X}{T} \right) \right]^{-1}$

$$J(\theta) = T \left(\frac{1}{T} X'(Y - Z\theta) \right)' \left(\frac{X'X}{T} \right)^{-1} \frac{1}{T} X'(Y - Z\theta)$$

$$J'' = \left(X'(Y - Z\theta) \right)' (X'X)^{-1} (X'(Y - Z\theta))$$

$$\frac{\partial J(\theta)}{\partial \theta} = 2 Z'X (X'X)^{-1} (X'(Y - Z\theta)) = 0$$

$$\hat{\theta} = \left[Z'X (X'X)^{-1} X'Z \right]^{-1} Z'X (X'X)^{-1} X'Y$$

In sum,

$$\sqrt{n}(\hat{\theta}_e - \theta) \xrightarrow{d} N(0, V_e)$$

$$V_e = \left(Z'X S_0^{-1} X'Z \right)^{-1}$$

$$\hat{V}_e = \left(\frac{1}{T} Z'X \left(\frac{X'X}{T} \right)^{-1} \frac{1}{T} X'Z \right)^{-1}$$

$$= T \left(Z'X (X'X)^{-1} X'Z \right)^{-1}$$

Remark

$$Y = Z\theta + u$$

$\Rightarrow$ premultiply X'

$$X'Y = X'Z\theta + X'u, \quad \text{cov}(X'u) = X'X$$

let $P = (X'X)^{-1} X' \Rightarrow P'X'Y = P'X'Z\theta + P'X'u$

Note $\text{cov}(P'X'u) = I$

$\Rightarrow$ RUN OLS,

$$\hat{\theta} = \left(Z'X P'P X'Z \right)^{-1} Z'X P'P X'Y$$

$$= \left[Z'X (X'X)^{-1} X'Z \right]^{-1} Z'X (X'X)^{-1} X'Y$$

Feasible Efficient GMM estimator

I) Two-Step Estimator

Step 1: Compute the initial consistent GMM estimator $\tilde{\theta}$ by setting $W = \left(\frac{X'X}{T} \right)^{-1}$ and compute $\tilde{u} = Y - Z\tilde{\theta}$. Compute $\hat{\Sigma}$ by using an appropriate method.

Step 2: Estimate the efficient GMM by using $\hat{\Sigma}$.

II. Continuous Updating Estimator

Statistical Inference

• Test of Overidentifying Moment Restriction.

$\Rightarrow$ We want to test that the extra $(k-p)$ moment conditions are zero or not

$$H_0: E(X_t' u_t) = 0 \quad \forall t, \quad H_1: H_0 \text{ is false}$$

Remark:

The H_1 says that at least one of the extra moment condition is false, though it does not specify which ones are taken as just identifying conditions.

$$J(\hat{\theta}) = T \left(\frac{1}{T} X'(Y - Z\hat{\theta}) \right)' \left(\frac{X'X}{T} \right)^{-1} \left(\frac{1}{T} X'(Y - Z\hat{\theta}) \right)$$

$$\xrightarrow[H_0]{d} \chi^2(k-p)$$

Wald-type test

LR-type test interpretation

- Test of a Subset of Overidentifying Moment Restriction

$$E(x_t u_t) = E \begin{pmatrix} x_t' u_t \\ x_t' u_t \end{pmatrix}$$

$$H_0: E(x_t' u_t) = 0 \quad H_1: H_0 \text{ is false}$$

1) Wald type Test

$$J_1(\hat{\theta}) = T \left(\frac{1}{T} x_1' (y_1 - z_1 \hat{\theta}) \right) W_e \left(\frac{1}{T} x_1' (y_1 - z_1 \hat{\theta}) \right)$$

$\downarrow$
 the corresponding optimal weight matrix

$$\xrightarrow[H_0]{d} \chi^2(k_1)$$

2) LR type Test

$$J_1(\hat{\theta}) - J_2(\hat{\theta}) \xrightarrow[H_0]{d} \chi^2(k_1)$$

$$T \left(\frac{1}{T} x_2' (y_2 - z_2 \hat{\theta}) \right) W_{22} \left(\frac{1}{T} x_2' (y_2 - z_2 \hat{\theta}) \right)$$

$\downarrow$
 the submatrix of the original optimal weighting matrix.

- Test of Linear Restrictions on Coefficient Parameter

$$\tilde{\theta} = \theta(\hat{\Omega}) - \tilde{V} R' (R \tilde{V} R')^{-1} (R \theta(\hat{\Omega}) - q)$$

where

$$\theta(\hat{\Omega}) = (z' x (x' \hat{\Omega} x)^{-1} x' z)' (z' x (x' \hat{\Omega} x)^{-1} x' y)$$

$$V = (z' x (x' \hat{\Omega} x)^{-1} x' z)^{-1}$$

$$\tilde{x} = (R \tilde{V} R')^{-1} (R \theta(\hat{\Omega}) - q)$$

$$\hookrightarrow \frac{1}{\sqrt{T}} \tilde{x} \xrightarrow[\text{under } H_0]{d} N(0, (R \tilde{V} R')^{-1})$$

1) Wald type test $\rightarrow$ no T here, used to estimate $\hat{V}$.

$$WT = (R \hat{\theta} - q)' (R \hat{V} R')^{-1} (R \hat{\theta} - q) \xrightarrow{d} \chi^2(r)$$

$$\text{where } \hat{V} = [z' x (x' \hat{\Omega} x)^{-1} x' z]^{-1}$$

2) LR-type test

$$LR = J_{\text{restricted}} - J_{\text{unrestricted}} \text{ (may not be +)}$$

$\downarrow$

$$LR = J_r(\tilde{\theta}_u) - J_u(\hat{\theta}) \rightarrow \chi^2(r) \text{ same as the unrestricted}$$

$$J_r(\tilde{\theta}_u) = T \left(\frac{1}{T} x' (y - z \tilde{\theta}_u) \right)' \left(\frac{x' \hat{\Omega} x}{T} \right)^{-1} \left(\frac{1}{T} x' (y - z \tilde{\theta}_u) \right)$$

$$\tilde{\theta}_u = \hat{\theta} - \hat{V} R' (R \hat{V} R')^{-1} (R \hat{\theta} - q)$$

3) LM see (p16)

Qualitative Response

$$y_t = \begin{cases} 1 & \text{if } y_t^* > 0 \\ 0 & \text{if } y_t^* \leq 0 \end{cases}$$

$$y_t^* = x_t \beta + u_t$$

$$u_{i1} - u_{i0} \quad z_{i1} - z_{i0} \quad \varepsilon_{i1} - \varepsilon_{i0}$$

$$E(y_t) = 1 \cdot P(y_t = 1) = 1 \cdot P(y_t^* > 0)$$

$$= 1 \cdot P(u_t > -x_t \beta) = 1 - F(-x_t \beta)$$

$$= F(x_t \beta) \text{ provided } f \text{ is symmetric}$$

Non-Constant Marginal Effects on Regression

$$\frac{\partial E(y_t)}{\partial x_{tk}} = f(x_t \beta) \beta_k \quad \text{on Probability}$$

the marginal effects on
the (conditional) marginal
density of variables

Marginal effect of a regressor on
odds ratio

$$q = \frac{P(y_t=1)}{P(y_t=0)} = \frac{F(x_t \beta)}{1 - F(x_t \beta)}$$

$$\frac{\partial q}{\partial x_{tk}} = \frac{f \beta_k (1 - F) - F(-f) \beta_k}{(1 - F)^2}$$

$$= \frac{f}{(1 - F)^2} \beta_k$$

Wondershare™

PDF Editor

QR-1

Estimation

I) Least Square (2011 1st round)

To be unbiased

$$E(u_t) = 0$$

"

$$\begin{aligned} E(y_t - x_t \beta) &= (1 - x_t \beta) P(y_t = 1) + (-x_t \beta) P(y_t = 0) \\ &= (1 - x_t \beta) P(y_t^* > 0) + (-x_t \beta) P(y_t^* \leq 0) \\ &= (1 - x_t \beta) F - x_t \beta (1 - F) = x_t \beta - F \end{aligned}$$

$$\text{Thus, } F = x_t \beta$$

$$E(u_t^2) = F(1 - F)$$

In sum, GLS is equivalent to
MLE under the assumption of the
standard uniform distribution.

II) MLE

$$L(y, \beta) = F(x_t \beta)^{y_t} (1 - F(x_t \beta))^{1 - y_t} \quad (y_t = 0, 1)$$

$$\Rightarrow L(y, \beta) = \prod_{t=1}^N F_t^{y_t} (1 - F_t)^{1 - y_t}$$

$$\Rightarrow \ell(y, \beta) = \sum_{t=1}^N y_t \ln F_t + (1 - y_t) \ln (1 - F_t)$$

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \Xi^{-1})$$

$$\text{where } \Xi = E \left(- \frac{1}{n} \frac{\partial^2 \ell(y, \beta)}{\partial \beta \partial \beta'} \right)$$

III) EM $\rightarrow$ maximization

$$\tilde{y}_t = \begin{cases} E(y_t^+) & \text{if } y_t = 1 \\ E(y_t^-) & \text{if } y_t = 0 \end{cases}$$

we usually impose $\sigma = 1$
 $\frac{\phi(x_t\beta)}{1 - \Phi(x_t\beta)}$

$$E(y_t^+) = E(y_t^+ | y_t^+ > 0)$$

$$= E(x_t\beta + u_t | u_t > -x_t\beta) = x_t\beta + E(u_t^+)$$

$$E(y_t^-) = E(y_t^- | y_t^- \leq 0) = x_t\beta + E(u_t^-)$$

Step 1: Run LPM y_t on x_t to get β'

Step 2: Compute $\tilde{y}$

Step 3: Run OLS $\tilde{y}$ on x_t to get β^2

Step 4: Repeat step 2-3 to get converged β

Drawback of EM:

The asymptotic covariance matrix is not a part of the computational procedure and has to be computed separately.

Remark:

3 tests: see p. 6.

Remark:

$$\hat{\beta}_{EM} \rightarrow \hat{\beta}_{MLE}$$

② Distribution of Estimators

(cov($\hat{\beta}$))

of Probability and Marginal Effects

$$a) F(x_t\beta) \sim N(F(x_t\beta), F^2 x_t V x_t)$$

$$b) \hat{\beta} \sim N(\beta, (f_t' \frac{\partial f}{\partial \beta} \beta x_t) V (f_t' \frac{\partial f}{\partial \beta} \beta x_t))$$

OQR

$$y_t = \begin{cases} 0 & \text{if } y_t^+ < d_1 \\ 1 & \text{if } d_1 \leq y_t^+ < d_2 \\ 2 & \text{if } y_t^+ \geq d_2 \end{cases}$$

$$p(y_t = 0) = p(x_t\beta + u_t < d_1) = F(d_1 - x_t\beta)$$

$$p(y_t = 1) = F(d_2 - x_t\beta) - F(d_1 - x_t\beta)$$

$$p(y_t = 2) = 1 - F(d_2 - x_t\beta)$$

$$L(y, \beta) = \prod_{y_t=0} F(d_1 - x_t\beta) \prod_{y_t=1} [F(d_2 - x_t\beta) - F(d_1 - x_t\beta)] \prod_{y_t=2} [1 - F(d_2 - x_t\beta)]$$

$$l(y, \beta) = \sum_{y_t=0} \ln F(d_1 - x_t\beta) + \sum_{y_t=1} \ln [F(d_2 - x_t\beta) - F(d_1 - x_t\beta)] + \sum_{y_t=2} \ln (1 - F(d_2 - x_t\beta))$$

$\hat{y} = \text{denom}$
 $L(y) = \prod_{t=1}^N \prod_{j=1}^J f_{t,j}^{y_{t,j}}$
 $y_{t,j} = \begin{cases} 1, & \text{if } j \text{ has been chosen} \\ 0, & \text{otherwise} \end{cases}$

$$\text{In general, } l(y, \beta) = \sum_{t=1}^N \sum_{j=1}^J \ln [F(d_{j+1} - x_t\beta) - F(d_j - x_t\beta)]$$

$$p_{ij} = F(d_{j+1} - x_t\beta) - F(d_j - x_t\beta)$$

$$d_{J+1} = \infty, d_0 = -\infty$$

$$\frac{\partial p_{ij}}{\partial x_t} = [f(d_{j+1} - x_t\beta) - f(d_j - x_t\beta)] (-\beta) = [f(d_j - x_t\beta) - f(d_{j+1} - x_t\beta)] \beta$$

QR-2 The sign of this marginal effect depends not only the sign of β , but also the sign of the difference b/w 2 values of the pdf.

UQR

multinomial: $\gamma = 0$ mixed $\beta \neq 0$
 Conditional: $\beta_j = 0$ $\gamma \neq 0$

→ individual specific (ex. income)

$$U_{ij} = X_{ij}\beta_j + Z_{ij}\gamma + \varepsilon_{ij} \equiv m_{ij} + \varepsilon_{ij}$$

↳ alternative specific (ex. price)

$$P_{ij} = P(Y_{ij}=1) = P(U_{ij} = \max_{k \in J} U_{ik})$$

$$= P(U_{ij} > U_{ik} \quad \forall k \neq j)$$

$$= P(m_{ij} + \varepsilon_{ij} > m_{ik} + \varepsilon_{ik} \quad \forall k \neq j)$$

$$= P(\varepsilon_{ik} < (m_{ij} - m_{ik}) + \varepsilon_{ij} \quad \forall k \neq j)$$

$$= \int P(\varepsilon_{ik} < m_{ij} - m_{ik} + \varepsilon_{ij} \mid \varepsilon_{ij}) f(\varepsilon_{ij}) d\varepsilon_{ij}$$

$$= \int \underbrace{\prod_{k \neq j} F(m_{ij} - m_{ik} + \varepsilon_{ij})}_{\Psi} f(\varepsilon_{ij}) d\varepsilon_{ij}$$

If $F \sim$ Gumbel, ie

$$F(z) = e^{-e^{-z}}, \quad f(z) = e^{-z} e^{-e^{-z}} = e^{-z} F'(z),$$

then.

$$P_{ij} = \frac{e^{m_{ij}}}{\sum_{k \in J} e^{m_{ik}}} = \frac{e^{X_{ij}\beta_j + Z_{ij}\gamma}}{\sum_{k \in J} e^{X_{ik}\beta_k + Z_{ik}\gamma}}$$

<Key>

$$\tau = \sum_{k \neq j} e^{-(m_{ij} - m_{ik})}$$

$$w = \varepsilon_{ij} - \ln(1 + \tau)$$

PF:

$$\text{Step 1: } \ln \Psi = -\varepsilon_{ij} - e^{-w}$$

$$w = \varepsilon_{ij} - \ln(1 + \tau), \quad \tau = \sum_{k \neq j} e^{-(m_{ij} - m_{ik})}$$

$$\Psi = \prod_{k \neq j} F(m_{ij} - m_{ik} - \varepsilon_{ij}) f(\varepsilon_{ij})$$

$$= \left(\prod_{k \neq j} e^{-e^{-(m_{ij} - m_{ik} - \varepsilon_{ij})}} \right) e^{-\varepsilon_{ij}} F(\varepsilon_{ij})$$

$$\ln \Psi = - \sum_{k \neq j} e^{-(m_{ij} - m_{ik} - \varepsilon_{ij})} - \varepsilon_{ij} - e^{-\varepsilon_{ij}}$$

$$= -e^{-\varepsilon_{ij}} \left(\underbrace{\sum_{k \neq j} e^{-(m_{ij} - m_{ik})}}_{\tau} + 1 \right) - \varepsilon_{ij}$$

$$= -\varepsilon_{ij} - e^{-\varepsilon_{ij}} + \ln(1 + \tau)$$

Step 2:

$$\Psi = \frac{1}{1 + \tau} e^{-w} e^{-e^{-w}}$$

$$\ln \Psi = -\varepsilon_{ij} - e^{-w}$$

$$= -w - \ln(1 + \tau) - e^{-w}$$

$$\therefore \Psi = e^{-w - \ln(1 + \tau) - e^{-w}}$$

$$= e^{-w} \frac{1}{1 + \tau} e^{-e^{-w}}$$

$$\therefore \Psi = \frac{1}{1 + \tau} e^{-w} e^{-e^{-w}}$$

Step 3:

$$P_{ij} = \int \Psi d\varepsilon_{ij} = \int \Psi dw = \frac{1}{1 + \tau}$$

Step 4:

$$1 + \tau = 1 + \sum_{k \neq j} e^{-(m_{ij} - m_{ik})}$$

$$p_{ij} = \frac{1}{1 + \tau} = \frac{1}{1 + e^{-m_{ij}} \sum_{k \neq j} e^{+m_{ik}}}$$

$$= \frac{1}{e^{-m_{ij}} (e^{m_{ij}} + \sum_{k \neq j} e^{m_{ik}})} = \frac{e^{m_{ij}}}{\sum_k e^{m_{ik}}}$$

Two coefficient vectors $\beta_j, \tilde{\beta}_j$ give an identical probability $p_{ij} = 1$

Note:

Parameters of individual-specific regressors x_i (ie β_j) are not all identifiable.

$$p_{ij} = \frac{e^{x_i \beta_j + z_{ij} \gamma}}{\sum_{k=1}^J e^{x_i \beta_k + z_{ik} \gamma}}$$

$$= \frac{e^{x_i \tilde{\beta}_j + z_{ij} \gamma}}{\sum_{k=1}^J e^{x_i \tilde{\beta}_k + z_{ik} \gamma}} \quad \text{where } \tilde{\beta}_j = \beta_j + \alpha$$

Thus, let $\beta_1 = 0$

$$p_{i1} = \frac{e^{z_{i1} \gamma}}{e^{z_{i1} \gamma} + \sum_{k=2}^J e^{x_i \beta_k + z_{ik} \gamma}}$$

$$p_{ij} = \frac{e^{z_{ij} \gamma}}{e^{z_{i1} \gamma} + \sum_{k=2}^J e^{x_i \beta_k + z_{ik} \gamma}}$$

Multinomial logit Model ($\gamma = 0$)

$$\frac{p_{ij}}{p_{i1}} = e^{x_i \beta_j}, \quad \frac{p_{ij}}{p_{ik}} = e^{x_i (\beta_j - \beta_k)}$$

odds ratio

$$\ln \left(\frac{p_{ij}}{p_{ik}} \right) = x_i (\beta_j - \beta_k)$$

$$\frac{\partial \ln \left(\frac{p_{ij}}{p_{ik}} \right)}{\partial x_i} = \beta_j - \beta_k$$

the differences in coefficients represent the marginal effects of regressors on the difference in log probabilities.

✓ Estimation (MLE)

$$L = \prod_{i=1}^N \prod_{j=1}^J p_{ij}^{y_{ij}}$$

$$l = \ln L = \sum_i \sum_j y_{ij} \ln p_{ij}$$

$$= \sum_i \left(-y_{i1} \ln D + \sum_{k=2}^J y_{ik} (x_i \beta_k - \ln D) \right)$$

$$= \sum_i \left(-\ln D + \sum_{k=2}^J y_{ik} x_i \beta_k \right)$$

the marginal effect of a particular alternative depend on all values of coefficient vectors.

$$\frac{\partial l}{\partial \beta_j} = \sum_i \left(-\frac{1}{D} e^{x_i \beta_j} x_i' + y_{ij} x_i' \right)$$

$$= \sum_i (y_{ij} - p_{ij}) x_i'$$

Wondershare™

PDF Editor QR-4

$$\frac{\partial^2 \ell}{\partial \beta \partial \beta'} = - \sum_{i=1}^n p_{ij} (1-p_{ij}) X_i' X_i$$

Estimation: (MLE)

$$L = \prod_{i=1}^N \prod_{j=1}^J p_{ij}^{y_{ij}}$$

Conditional Logit Model: ($\beta=0$)

$$p_{ij} = \frac{e^{z_{ij} \delta}}{\sum_{k=1}^J e^{z_{ik} \delta}}$$

$$\frac{p_{ij}}{p_{ik}} = e^{(z_{ij} - z_{ik}) \delta}$$

$$\Rightarrow \ln\left(\frac{p_{ij}}{p_{ik}}\right) = (z_{ij} - z_{ik}) \delta$$

$$\begin{aligned} \ell = \ln L &= \sum_i \sum_j y_{ij} \ln p_{ij} \\ &= \sum_i \sum_j y_{ij} (z_{ij} \delta - \ln D) \end{aligned}$$

$$\begin{aligned} \frac{\partial \ell}{\partial \delta} &= \sum_i \sum_j y_{ij} \left(z_{ij}' - \frac{1}{D} \sum_j e^{z_{ij} \delta} z_{ij}' \right) \\ &= \sum_i \sum_j y_{ij} (z_{ij}' - \bar{z}_i') \end{aligned}$$

Independence of Irrelevant Alternatives

Log odds ratio b/w two alternatives depends on δ only the attributes of alternatives in comparison.

$$\frac{\partial^2 \ell}{\partial \delta \partial \delta'} = - \sum_i \sum_j y_{ij} p_{ik} (z_{ik} - \bar{z}_i') (z_{ik} - \bar{z}_i')$$

↓
try to calculate again.

$$\frac{\partial p_{ij}}{\partial z_{ij}} = \frac{1}{D^2} \left(e^{z_{ij} \delta} \delta' D - e^{z_{ij} \delta} e^{z_{ij} \delta} \delta' \right)$$

$$= p_{ij} (1-p_{ij}) \delta'$$

$$\begin{aligned} \frac{\partial p_{ij}}{\partial z_{is}} &= \frac{1}{D^2} \left(-e^{z_{ij} \delta} e^{z_{is} \delta} \delta' \right) \\ &= -p_{ij} p_{is} \delta' \end{aligned}$$

The marginal effect of an alternative-specific attribute on its own probability has the same as the sign of the coefficient δ' , while the cross marginal effect has a sign opposite to the sign of δ' .

Nested Logit Model

$$U_{jk} = m_{jk} + \varepsilon_{jk}, \quad j=1,2, \quad k = \begin{cases} 1 & \text{if } j=1 \\ 1,2 & \text{if } j=2 \end{cases}$$

group alternative

Suppose

$$F(\varepsilon_{11}) = e^{-e^{-\varepsilon_{11}}} \text{ and}$$

$$F(\varepsilon_{21}, \varepsilon_{22}) = e^{-\left[e^{-\frac{\varepsilon_{21}}{\rho}} + e^{-\frac{\varepsilon_{22}}{\rho}} \right]^\rho}, \quad \rho \in [0,1]$$

Remark:

$\rho=1$ implies the independence of ε_{21} and ε_{22}

Then,

$$e^{m_{11}}$$

$$P_{11} = \frac{e^{m_{11}}}{e^{m_{11}} + \left[e^{\frac{m_{21}}{\rho}} + e^{\frac{m_{22}}{\rho}} \right]^\rho}$$

$$P_{k|2} = \frac{e^{\frac{m_{2k}}{\rho}}}{e^{\frac{m_{21}}{\rho}} + e^{\frac{m_{22}}{\rho}}}, \quad k=1,2$$

Remark:

P_{11} - the choice b/w the car and the bus

$P_{k|2}$: given choosing 2 (ie bus), the choice b/w $k=1$ and $k=2$

Then,

$$L^i = P_{11} \cdot P_2 (P_{1|2} P_{2|2})$$

$$\Rightarrow \ln L^i = (\ln P_{11} + \ln P_2) + (\ln P_{1|2} + \ln P_{2|2})$$

Alternative 2-step Estimation.

See P15

Censored Dependent Variable

$$y_t = \begin{cases} y_t^*, & y_t^* > 0 \\ 0, & y_t^* \leq 0 \end{cases}$$

$$\begin{aligned} E(y_t^+) &= E(y_t^* | y_t^* > 0) \\ &= E(x_t\beta + u_t | u_t > -x_t\beta) \\ &= x_t\beta + E(u_t | u_t > -x_t\beta) \\ &= x_t\beta + \sigma \frac{\phi(\frac{x_t\beta}{\sigma})}{\Phi(\frac{x_t\beta}{\sigma})} \quad \text{OLS is biased} \end{aligned}$$

$$\begin{aligned} E(y_t) &= E(y_t | y_t > 0)P(y_t > 0) \\ &\quad + E(y_t | y_t = 0)P(y_t = 0) \\ &= E(y_t^* | y_t^* > 0)P(y_t^* > 0) \\ &= P_t^+ E(y_t^+) \quad \text{II } 1 - F(-x_t\beta) = F(x_t\beta) \end{aligned}$$

$$E(y_t^*) = E(x_t\beta + u_t) = x_t\beta$$

Marginal effects of Regressors

$$\frac{\partial E(y_t^*)}{\partial x_{tk}} = \beta_k \quad \begin{array}{l} \text{the effect} \\ \text{of the regressor} \\ \text{on the change} \\ \text{of households} \\ \text{which have a} \\ \text{positive expenditure} \end{array} \quad \begin{array}{l} \text{the effect} \\ \text{of the regressor} \\ \text{on the expenditures} \\ \text{of those who had} \\ \text{positive expenditures} \end{array}$$

$$\frac{\partial E(y_t)}{\partial x_{tk}} = \frac{\partial P_t^+}{\partial x_{tk}} E(y_t^+) + P_t^+ \frac{\partial E(y_t^+)}{\partial x_{tk}}$$

$$\frac{\partial E(y_t^*)}{\partial x_{tk}} = \beta_k + \frac{\partial E(u_t^+)}{\partial x_{tk}}$$

$$\frac{\partial P_t^+}{\partial x_{tk}} = -f(-x_t\beta)(-\beta_k) = f(x_t\beta)\beta_k$$

Estimation

I) MLE

$$d_t = \begin{cases} 0, & y_t = 0 \quad y_t^* \leq 0 \\ 1, & y_t > 0, \quad y_t^* > 0 \end{cases}$$

$$P(d_t = 0) = P(y_t^* \leq 0) = P(u_t \leq -x_t\beta)$$

thus,

$$L = \prod_{i=1}^n F(-x_i\beta)^{1-d_i} f(y_i - x_i\beta)^{d_i}$$

$$l = \sum_{i=1}^n (1-d_i) \ln F(-x_i\beta) + d_i \ln f(y_i - x_i\beta)$$

Case: Tobit model

$$P(d_t = 0) = P(y_t^* \leq 0) = P(u_t \leq -x_t\beta)$$

$$= P\left(\frac{u_t}{\sigma} \leq \frac{-x_t\beta}{\sigma}\right) = \Phi\left(\frac{-x_t\beta}{\sigma}\right) = 1 - \Phi\left(\frac{x_t\beta}{\sigma}\right)$$

Thus,

$$l = \sum_{i=1}^n (1-d_i) \ln(1 - \Phi_t) \quad \text{not global concave}$$

$$+ d_t \left[\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma^2 - \frac{1}{2} \left(\frac{y_t - x_t\beta}{\sigma} \right)^2 \right]$$

$$\text{let } \tau = \frac{1}{\sigma}, \quad \alpha = \frac{\beta}{\sigma}$$

$$l = \sum_{i=1}^n \left[(1-d_i) \ln(1 - \Phi_t) - \Phi(x_t\alpha) + d_t \left(\frac{1}{2} \ln 2\pi + \ln \tau - \frac{1}{2} (\tau y_t - x_t\alpha)^2 \right) \right]$$

global concave

$$\frac{\partial l}{\partial \alpha} = \sum_{i=1}^N (1-d_t) \frac{-\phi}{1-\Phi} X_t' + d_t \left(-\frac{1}{2}\right) \cdot 2(\tau y_t - y_t d) X_t'$$

$$= \sum_{i=1}^N -(1-d_t) \lambda_t X_t' + d_t (\tau y_t - y_t d) X_t'$$

$$\frac{\partial l}{\partial \tau} = \sum_{i=1}^N d_t \frac{1}{\tau} - \frac{d_t}{2} \cdot 2(\tau y_t - y_t d) y_t$$

$$= \sum_i \frac{d_t}{\tau} - d_t (\tau y_t - y_t d) y_t$$

$$\frac{\partial^2 l}{\partial \alpha \partial \alpha'} = \sum_{i=1}^N d_t X_t' X_t$$

$$\frac{\partial^2 l}{\partial \alpha \partial \alpha'} = \sum_{i=1}^N -(1-d_t) X_t' (\lambda (\lambda - X_t d)) X_t - d_t X_t' X_t$$

$$\frac{\partial^2 l}{\partial \tau^2} = \sum_i -\frac{d_t}{\tau^2} - d_t y_t^2$$

$$\Rightarrow H = - \begin{pmatrix} \sum d_t X_t' X_t & \sum d_t X_t' y_t \\ \sum d_t y_t X_t & \sum d_t y_t^2 \end{pmatrix} - \begin{pmatrix} \sum (1-d_t) X_t' \lambda (\lambda - X_t d) X_t & 0 \\ 0 & \sum \frac{d_t}{\tau^2} \end{pmatrix}$$

$$= - \begin{pmatrix} X_2' X_2 & X_2' y_2 \\ y_2' X_2 & y_2' y_2 \end{pmatrix} - \begin{pmatrix} X_1' D X_1 & 0 \\ 0 & N_2 / \tau^2 \end{pmatrix}$$

$\Rightarrow H$ is nsd

$\Rightarrow$ log-likelihood fn is concave in α and τ

 **wondershare**™

PDF Editor CDV-2

II) EM

Case 1. β, σ^2

$$\tilde{y}_t^* = \begin{cases} y_t^* & \text{if } y_t^* > 0 \\ E(y_t^*) & \text{if } y_t^* \leq 0 \end{cases}$$

$$E(y_t^*) = E(y_t^* | y_t^* \leq 0)$$

$$= E(X_t \beta + u_t | u_t \leq -X_t \beta)$$

$$= X_t \beta - \sigma \frac{\phi(\frac{X_t \beta}{\sigma})}{\Phi(-\frac{X_t \beta}{\sigma})} = X_t \beta - \sigma \frac{\phi(\frac{X_t \beta}{\sigma})}{1 - \Phi(\frac{X_t \beta}{\sigma})}$$

Step 1: Choosing initial consistent estimators of $\hat{\beta}^{(0)}$ and $\hat{\sigma}^{(0)}$

Step 2: Compute $\tilde{y}_t$

Step 3: RUN OLS $\tilde{y}_t$ on X_t to get $\hat{\beta}^{(2)}$

Step 4: Compute $\hat{\sigma}^{(2)}$

$$\text{by } \hat{\sigma}^2 = \frac{\sum d_t (y_t - X_t \hat{\beta})^2 y_t}{\sum d_t}$$

Step 5: Repeat Step 2-4 until it converges

from p. 4

Case 2: τ, α

$$\tilde{y}_t = \begin{cases} \tau y_t^* & \text{if } y_t^* > 0 \\ \tau E(y_t^*) & \text{if } y_t^* < 0 \end{cases}$$

$$\tau E(y_t^*) = x_t \alpha - \frac{\phi(x_t \alpha)}{1 - \Phi(x_t \alpha)}$$

Step 1: For any initial $\hat{\tau}^1$ and $\hat{\alpha}^1$

Run OLS $\tilde{y}_t$ on x_t to

get $\hat{\alpha}^2$

Step 2: Compute $\hat{\tau}^2$ by

$$\left(\sum_{N_2} y_t^{*2} \right) \tau^2 - \left(\sum_{N_2} (x_t \hat{\alpha}) y_t^* \right) \tau - N_2 = 0$$

Step 3: Repeat step 1-2 until converges

III) Heckman's 2-steps Method

Step 1. Run Probit Model

$$\tilde{y}_t = \begin{cases} 1 & \text{if } y_t^* > 0 \\ 0 & \text{if } y_t^* < 0 \end{cases} \text{ on } x_t$$

to get $\tilde{\beta}$ and $\tilde{\sigma}$ only observed data

Step 2. Run OLS y_t^+ on x_t

$$\frac{\phi(x_t \tilde{\beta})}{\tilde{\sigma}}$$

and

$$\frac{\phi(x_t \tilde{\beta})}{\tilde{\sigma}}$$

including zero

or RUN OLS y_t^+ on $\frac{\phi(x_t \tilde{\beta})}{\tilde{\sigma}}$ and $\phi(x_t \tilde{\beta})$

OLS is biased and inconsistent

• Truncated Model

This is the case where observations on both dependent and independent variables are not available for a sub-population.

$$\begin{cases} E(u | u > -x\beta) = \sigma K, & K = \frac{\phi(\frac{x\beta}{\sigma})}{\Phi(\frac{x\beta}{\sigma})} \\ \text{Var}(u | u > -x\beta) = \sigma^2 (1 - K(K + x\alpha)) \end{cases}$$

Note

when $c < 0$, this $\frac{c}{\sigma}$ won't be affected.

$$E(u | u > c) = \sigma \frac{\phi(\frac{c}{\sigma})}{1 - \Phi(\frac{c}{\sigma})}$$

$$\text{Var}(u | u > c) = \sigma^2 (1 - K(K - \frac{c}{\sigma}))$$

The LSE of β is biased and inconsistent because of the nonzero mean of the error term.

The regression error term is also heteroskedasticity

Estimation (MLE)

$$f(y_t^* | y_t^* > 0) = \frac{\frac{1}{\sigma} \phi(\frac{y_t^* - x_t \beta}{\sigma})}{\Phi(\frac{x_t \beta}{\sigma})} \text{ if } y_t^* > 0$$

$$L = \sum_{t=1}^N \left[f(y_t^* | y_t^* > 0) \right] \text{ only for those } y_t^* > 0 \text{ (ie } N_2)$$

$$l = \sum_{t=1}^N \ln(f(y_t^* | y_t^* > 0))$$

$$= \sum_{t=1}^N \left[-\ln \Phi_t + \ln \frac{1}{\sigma} - \frac{1}{2} \ln a \pi - \frac{1}{2\sigma^2} (y_t^* - x_t \beta)^2 \right]$$

$$= \sum_{t=1}^N -\ln \Phi_t - \frac{N_2}{2} \ln \sigma^2 - \frac{N_2}{2} \ln a \pi$$

$$- \sum_{t=1}^N \frac{1}{2\sigma^2} (y_t^* - x_t \beta)^2$$

• Type II Tobit Model

$$y_{1t} = \begin{cases} y_{1t}^* & \text{if } y_{2t}^* > 0 \\ - & \text{if } y_{2t}^* \leq 0 \end{cases}$$

OLS

$$E(y_1) = E(y_1^* | y_2^* > 0)$$

$$= E(x_1\beta_1 + u_1 | x_2\beta_2 + u_2 > 0)$$

$$= x_1\beta_1 + E(u_1 | u_2 > -x_2\beta_2)$$

$$= x_1\beta_1 + E\left(\frac{\sigma_{12}}{\sigma_{22}} u_2 + \eta_1 \mid u_2 > -x_2\beta_2\right)$$

$$= x_1\beta_1 + \frac{\sigma_{12}}{\sigma_{22}} E(u_2 | u_2 > -x_2\beta_2)$$

$$= x_1\beta_1 + \frac{\sigma_{12}}{\sigma_{22}} \sqrt{\sigma_{22}} \frac{\phi\left(\frac{x_2\beta_2}{\sqrt{\sigma_{22}}}\right)}{\Phi\left(\frac{x_2\beta_2}{\sqrt{\sigma_{22}}}\right)} = x_1\beta_1 + \frac{\sigma_{12}}{\sqrt{\sigma_{22}}} K_2$$

$$\text{Var}(y_1) = \text{Var}(x_1\beta_1 + u_1 | u_2 > -x_2\beta_2)$$

$$= \text{Var}\left(\frac{\sigma_{12}}{\sigma_{22}} u_2 + \eta_1 \mid u_2 > -x_2\beta_2\right)$$

$$= \frac{\sigma_{12}^2}{\sigma_{22}^2} \text{Var}(u_2 | u_2 > -x_2\beta_2) + \text{Var}(\eta_1)$$

$$= \frac{\sigma_{12}^2}{\sigma_{22}^2} \sigma_{22} (1 - K_2(K_2 + x_2\beta_2)) + \sigma_{11} - \sigma_{12}\sigma_{22}^{-1}\sigma_{21}$$

$$= \sigma_{11} - \frac{\sigma_{12}^2}{\sigma_{22}} K_2(K_2 + x_2\beta_2)$$

$$\Rightarrow y_1 = x_1\beta_1 + \underbrace{\frac{\sigma_{12}}{\sqrt{\sigma_{22}}} K_2}_{\text{bias}} + \tilde{z}_1 \quad (*)$$

$$E(\tilde{z}_1) = 0 \quad E(\tilde{z}_1^2) = \sigma_{11} - \frac{\sigma_{12}^2}{\sigma_{22}} K_2(K_2 + x_2\beta_2)$$

the source of the
bias and inconsistency
of the least squares
estimator of β_1

Estimation

I) Heckman's two-step estimator

From (*),

$$y_1 = x_1\beta_1 + \frac{\sigma_{12}}{\sqrt{\sigma_{22}}} \frac{\phi(x_2\beta_2)}{\Phi(x_2\beta_2)}$$

Step 1: estimate d_2 w/ probit MLE.

(y_2 on x_2), (with setting $\sigma_{22}=1$)

Step 2: RUN OLS or GLS

y_1 on x_1 and K

II) MLE

$$y_i = \begin{cases} y_i^* & \text{if } y_2 = 1 \\ - & \text{if } y_2 = 0 \end{cases}$$

$$y_2 = \begin{cases} 1 & \text{if } y_2^* \geq 0 \\ 0 & \text{if } y_2^* < 0 \end{cases}$$

$$L = \prod_{y_2=0} P(y_2^* \leq 0) \prod_{y_2=1} P(y_2^* \geq 0) f(y_1^* | y_2^* \geq 0)$$

$$= \prod_{t=1}^N P(y_2^* \leq 0)^{1-y_2} \left[P(y_2^* \geq 0) f(y_1^* | y_2^* \geq 0) \right]^{y_2}$$

$$\textcircled{1} P(y_2^* \leq 0) = P(u_2 \leq -x_2\beta_2) = 1 - \Phi(x_2\beta_2)$$

$$\textcircled{2} P(y_2^* \geq 0) f(y_1^* | y_2^* \geq 0)$$

$$= P(y_2^* \geq 0 | y_1^*) f(y_1^*)$$

Remark: see p13

1) If $\sigma_{12} = 0$, then OLS of β_1 from $y_1 = x_1\beta_1 + \varepsilon_1$ is BLUE

2)

$$a) f(y_1^*) = \frac{1}{\sigma_1} \phi\left(\frac{y_1^* - x_1 \beta_1}{\sigma_1}\right) \quad u_2 \leq -y_2 \beta_2$$

$$b) p(y_2^* \geq 0 | y_1^*) = 1 - p(y_2^* < 0 | y_1^*)$$

$$= \Phi\left(\frac{x_2 \beta_2 + \frac{\sigma_{12}}{\sigma_{11}} (y_1 - x_1 \beta_1)}{\sqrt{\sigma_{22} - \sigma_{21} \sigma_{11}^{-1} \sigma_{12}}}\right)$$

$$= \Phi\left(\frac{x_2 d_2 - \frac{\rho}{\sqrt{\sigma_{11}}} (y_1 - x_1 \beta_1)}{(1 - \rho^2)^{1/2}}\right)$$

Note:

$$y_2 | y_1 \sim N\left(x_2 \beta_2 + \frac{\sigma_{12}}{\sigma_{11}} (y_1 - x_1 \beta_1), \sigma_{22} - \sigma_{21} \sigma_{11}^{-1} \sigma_{12}\right)$$

$$u_1 | u_2 \sim N\left(\frac{\sigma_{12}}{\sigma_{22}} (u_2), \sigma_{11} - \sigma_{12} \sigma_{22}^{-1} \sigma_{21}\right)$$

Thus,

$$L = \prod_{i=1}^N (1 - \Phi)^{1 - y_i} \left(\frac{1}{\sigma_1} \phi\left(\frac{y_i - x_i \beta_1}{\sigma_1}\right) \Phi\left(\frac{x_i d_2 - \frac{\rho}{\sqrt{\sigma_{11}}} (y_i - x_i \beta_1)}{(1 - \rho^2)^{1/2}}\right) \right)^{y_i}$$

$$\ln L = \sum_{i=1}^N (1 - y_i) \ln(1 - \Phi) + y_i \left[\ln \frac{1}{\sigma_1} \phi\left(\frac{y_i - x_i \beta_1}{\sigma_1}\right) + \ln \Phi\left(\frac{x_i d_2 - \frac{\rho}{\sqrt{\sigma_{11}}} (y_i - x_i \beta_1)}{(1 - \rho^2)^{1/2}}\right) \right]$$

Remark:

Since σ_2 appears in the likelihood function only in d_2 and ρ , σ_2 is not identifiable, hence can be set to $\sigma_2 = 1$.

Nonspherical Covariance of Error Term

~~(A3) $E(uu'|x) = \sigma^2 I$~~

(A3)* $E(uu'|x) = \sigma^2 \Omega$ symmetric and PSD
 $\text{tr}(\Omega) = N$

Effects of violating (A3)

↳ for the unbiasedness of $\hat{\sigma}_{GLS}^2$

1. $\hat{\beta} = (x'x)^{-1}x'y$ — linear ✓
2. $E(\hat{\beta}|x) = E(\beta + (x'x)^{-1}x'u|x) = \beta$ — unbiased ✓

$$\begin{aligned} 3. \text{Cov}(\hat{\beta}|x) &= E[(\hat{\beta} - E(\hat{\beta}))(\hat{\beta} - E(\hat{\beta}))'] \\ &= E[(x'x)^{-1}x'u u'x(x'x)^{-1}|x] \\ &= (x'x)^{-1}x'\Omega x(x'x)^{-1} = \sigma^2(x'x)^{-1} \end{aligned}$$

(not best any more)

$$\begin{aligned} 4. E(\hat{\sigma}^2) &= E\left(\frac{\hat{u}'\hat{u}}{N-k}\right) = \frac{E(u'u)}{N-k} = \frac{E(\text{tr}(u u'))}{N-k} \\ &= \frac{\text{tr}(M E(u u'))}{N-k} = \frac{\text{tr}(M \sigma^2 \Omega)}{N-k} = \sigma^2 \frac{\text{tr}(M \Omega)}{N-k} \neq \sigma^2 \end{aligned}$$

⇒ $\hat{\sigma}^2$ is biased.

Thm

$$\hat{\beta}_{GLS} = (x'\Omega^{-1}x)^{-1}x'\Omega^{-1}y \text{ is BLUE}$$

$$\text{where } \hat{\beta}_{GLS} \in \arg \min_{\beta} (y - x\beta)' \Omega^{-1} (y - x\beta)$$

Properties

$$\begin{aligned} 1. \text{Cov}(\hat{\beta}_{GLS}) &= \text{Cov}(\beta + (x'\Omega^{-1}x)^{-1}x'\Omega^{-1}u|x) \\ &= (x'\Omega^{-1}x)^{-1}x'\Omega^{-1}E(uu'|x)\Omega^{-1}x(x'\Omega^{-1}x)^{-1} \\ &= \sigma^2(x'\Omega^{-1}x)^{-1} \text{ (Best)} \end{aligned}$$

$$2. \hat{\sigma}^2 = \frac{(y - x\hat{\beta}_{GLS})'\Omega^{-1}(y - x\hat{\beta}_{GLS})}{N-k} \text{ is unbiased.}$$

PF (of the thm)

$$\min (y - x\beta)' \Omega^{-1} (y - x\beta)$$

$$\text{let } \Omega^{-1} = P'P$$

$$\Rightarrow \min (P(y - x\beta))'(P(y - x\beta))$$

Note

$$P(y - x\beta) \equiv u^*$$

$$\begin{aligned} \text{Cov}(u^*) &= \text{Cov}(P \cdot u) = P \cdot \text{Cov}(u) P' \\ &= P \cdot \sigma^2 \Omega P' = \sigma^2 P (P'P)^{-1} P' = \sigma^2 I \end{aligned}$$

Examples : P2 - P11

— AR(1)

— AR(2)

— MA(1)

★ SUR

↳ no intertemporal correlation

<Idea> strict exogeneity

$$E(x_{it} u_{is}) = 0 \quad \forall t, s$$

— Time Series - Cross Section Data (P5)

— FE

★ RE (quasi-demean)

$$Z = \begin{bmatrix} \sigma_a^2 + \sigma_u^2 & \sigma_a^2 & \dots & \sigma_a^2 \\ \sigma_a^2 & & & \\ & & \sigma_a^2 + \sigma_u^2 & \\ & & & \sigma_a^2 + \sigma_u^2 \end{bmatrix} \Rightarrow Q'Q = Z^{-1}$$

$$Q = \frac{1}{\sigma_u} (I - \delta e(e'e)^{-1}e'), \quad \delta = 1 - \frac{\sigma_u}{\sqrt{T\sigma_a^2 + \sigma_u^2}}$$

— ARCH (P8)

wondershare™

PDF Editor